

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Digital IO Connector Removal & Replacement	Effective Date	06 July 2017

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	06 July 2017	First Release	Chin Chee Yan

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Removal Procedure

1. Unload board from machine if any.
2. Go to X ray source tab and toggle “Turn off X-Ray”. Power off at Operator Console Panel

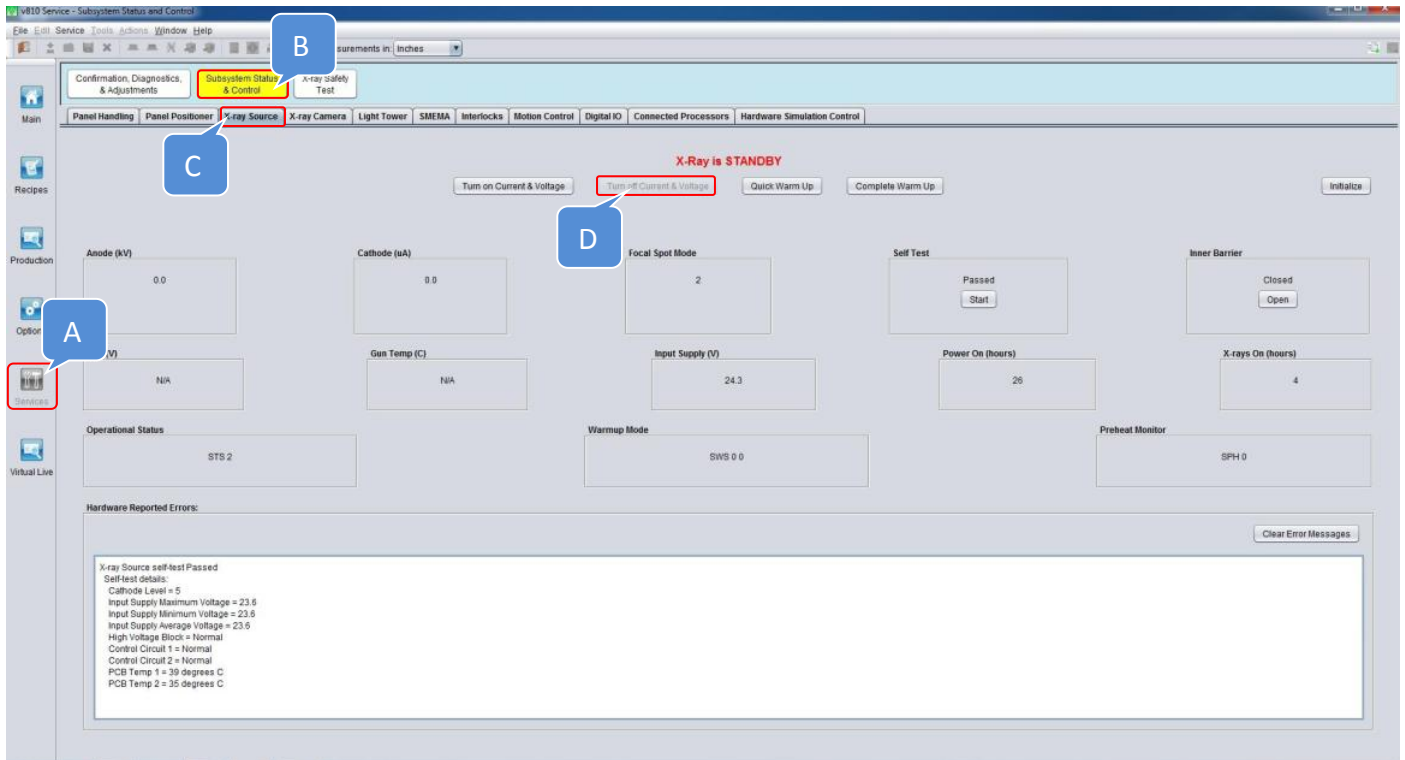


Figure 1 X Ray Source Shutdown

3. Power off SSAC

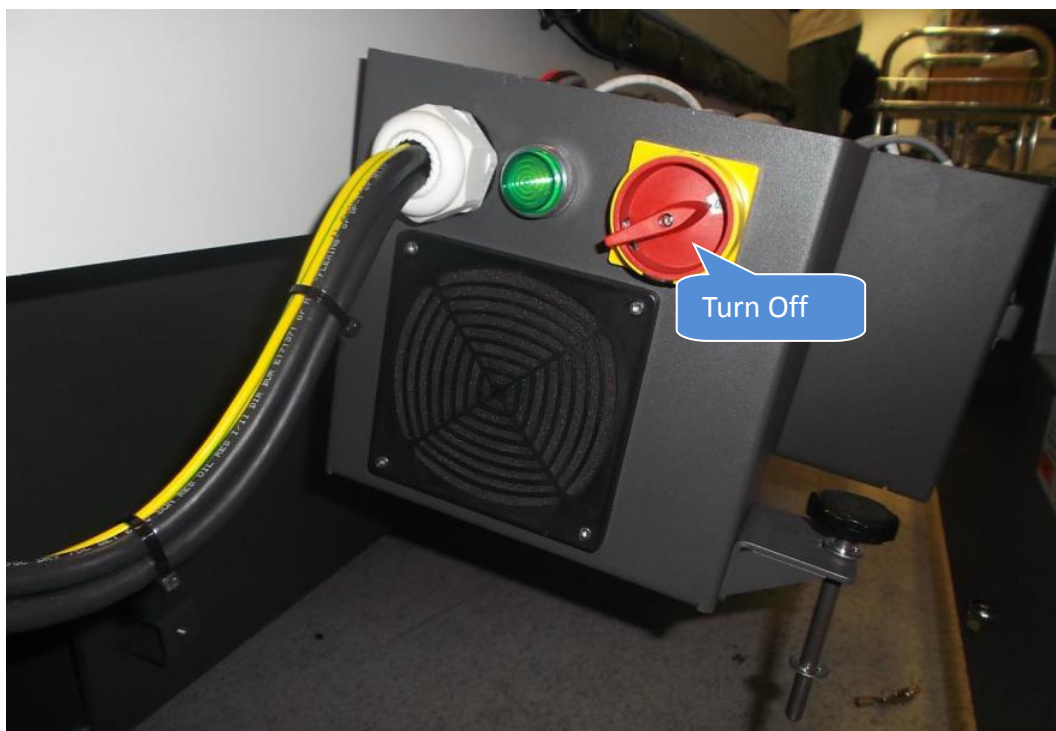


Figure 2 SSAC Power Off

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4. Gently loosen Both Anchors as below Figure 3

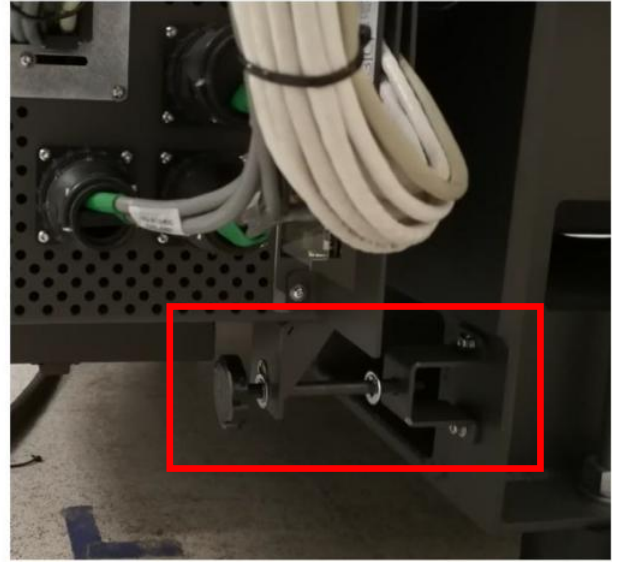
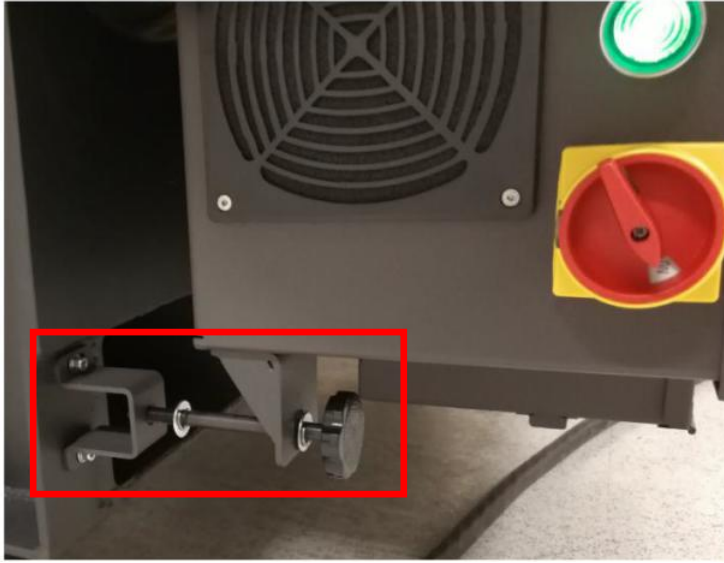


Figure 3 Anchors Release

5. Extend SSCA Support Rod which locate underneath of SSCA as below Figure 4



Figure 4 SSCA Support Rod Extend

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I/O, MODULE, SYNQNET NETWORK ADAPTER Connector Removal

- Probe Tiny Square Button with Flat Head Screwdriver to disconnect IO Cables.

Remarks : Perform Labeling at IO Cable upon removal. Applicable only for Cable Without Label

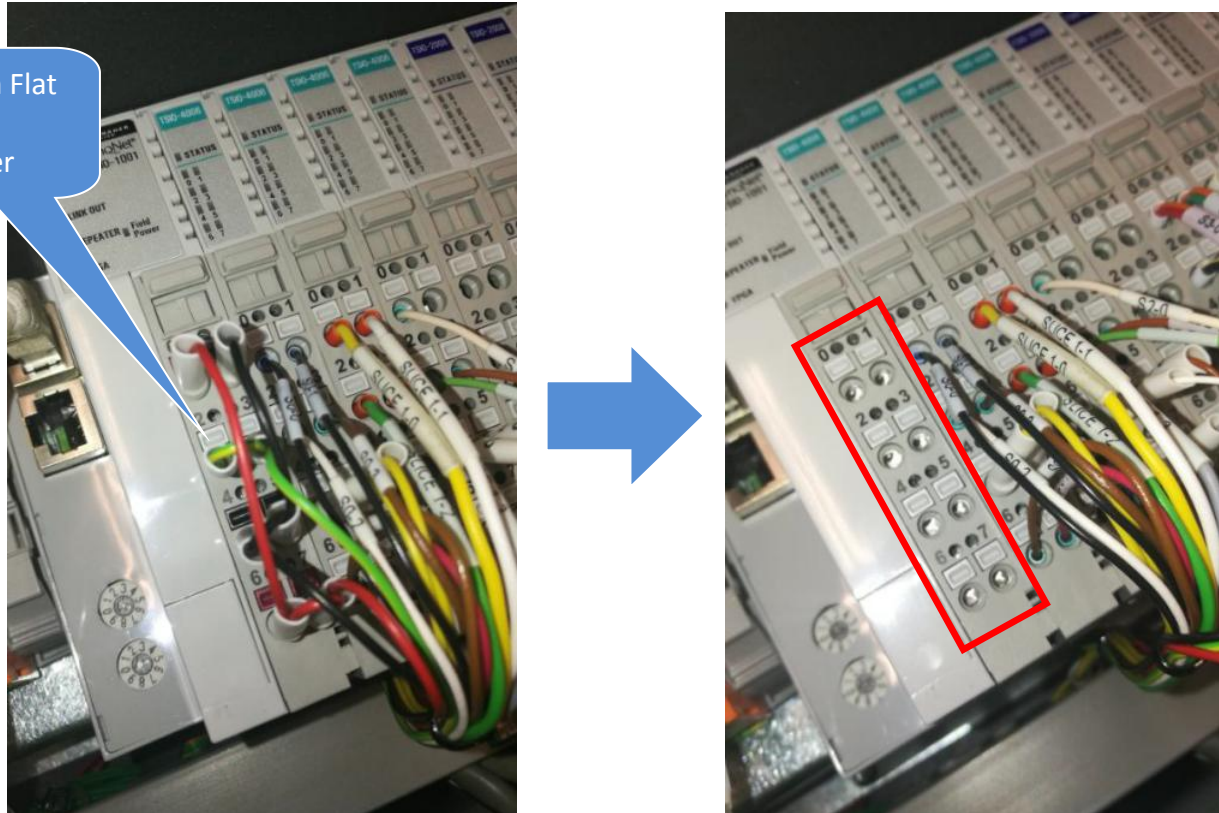


Figure 5 IO Cable Disconnect

- Gently Pull out IO Connector Hinge. Lift up and remove the IO Connector from Slice IO SYNQNET NETWORK ADAPTER as below Figure 6

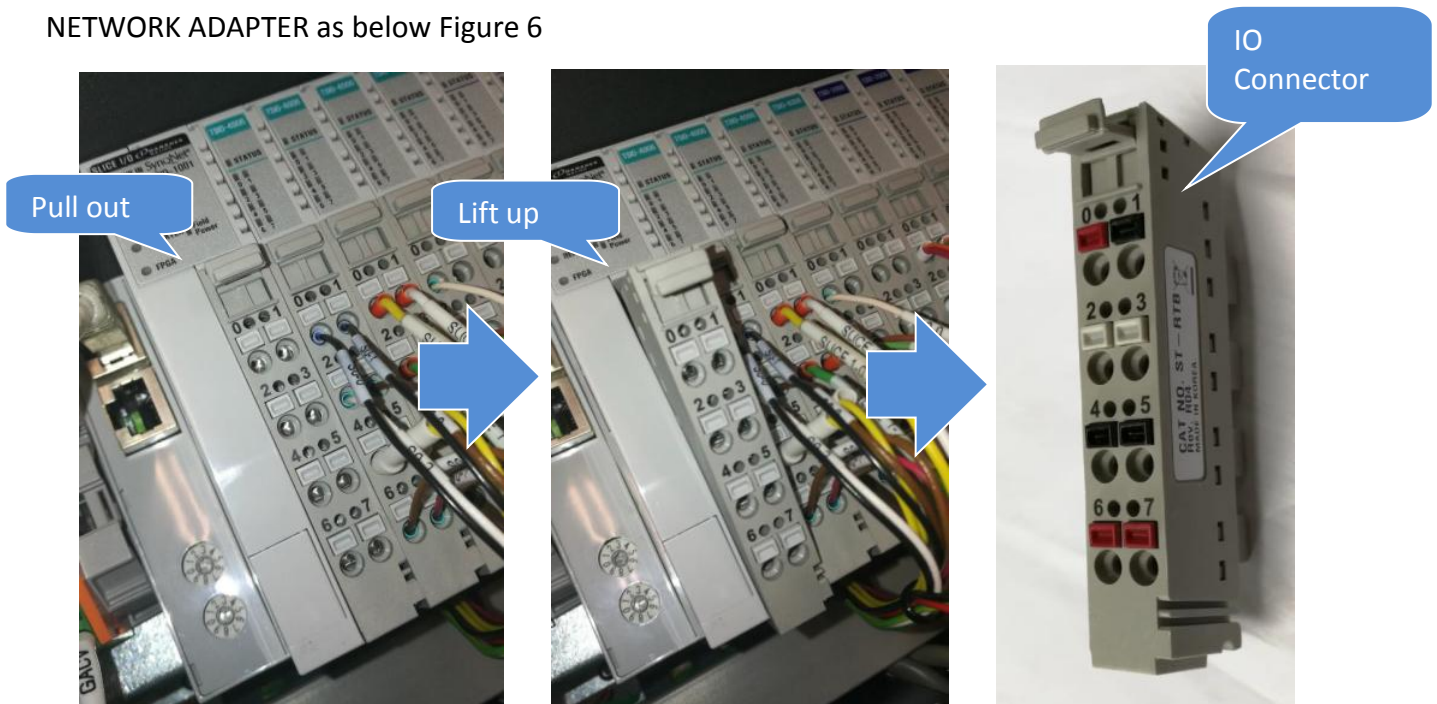


Figure 6 Slice IO Connector (SYNQNET NETWORK ADAPTER)

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Selected Slice IO Connector Removal

8. Gently probe the Tiny Square Button at Selected Slice IO. Disconnect all IO Cable from Selected Slice IO

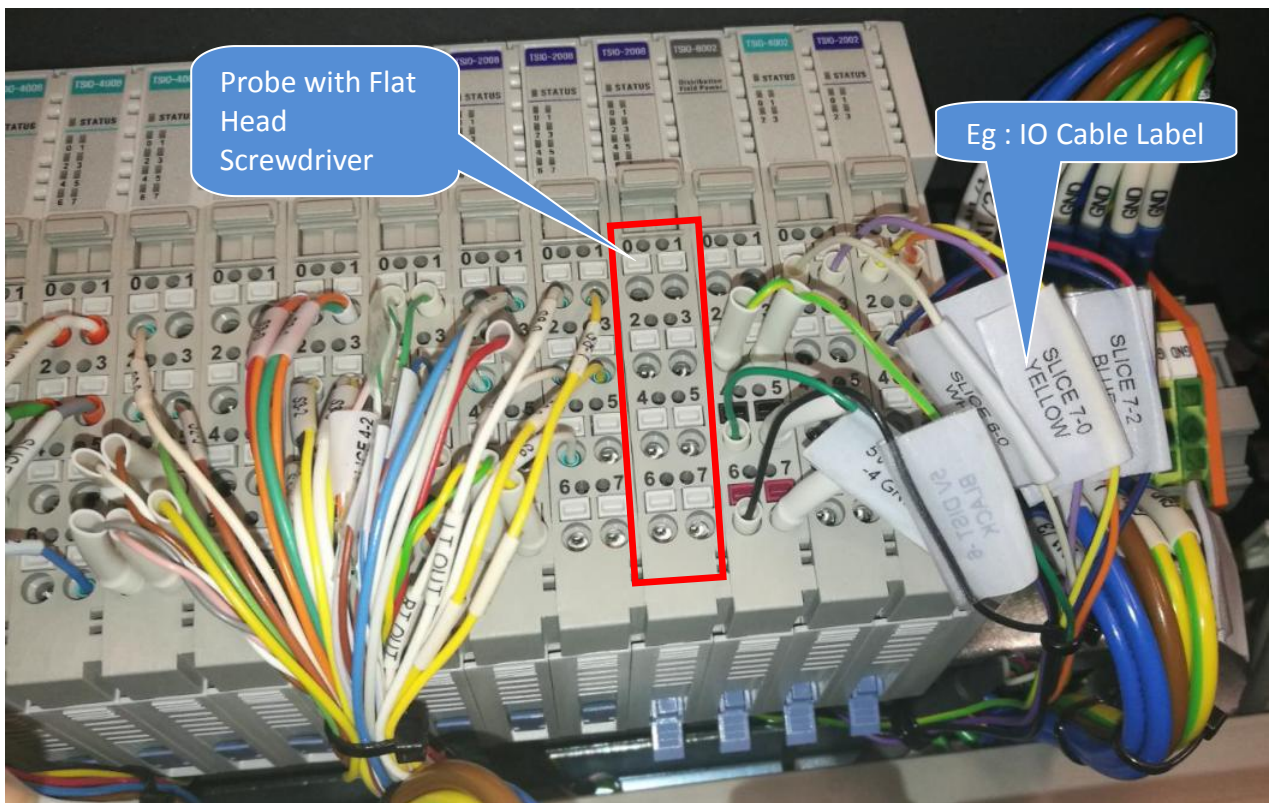


Figure 7 Selected IO Cable Removal

9. Gently Pull out IO Connector Hinge. Lift up and remove the IO Connector from Selected Slice IO

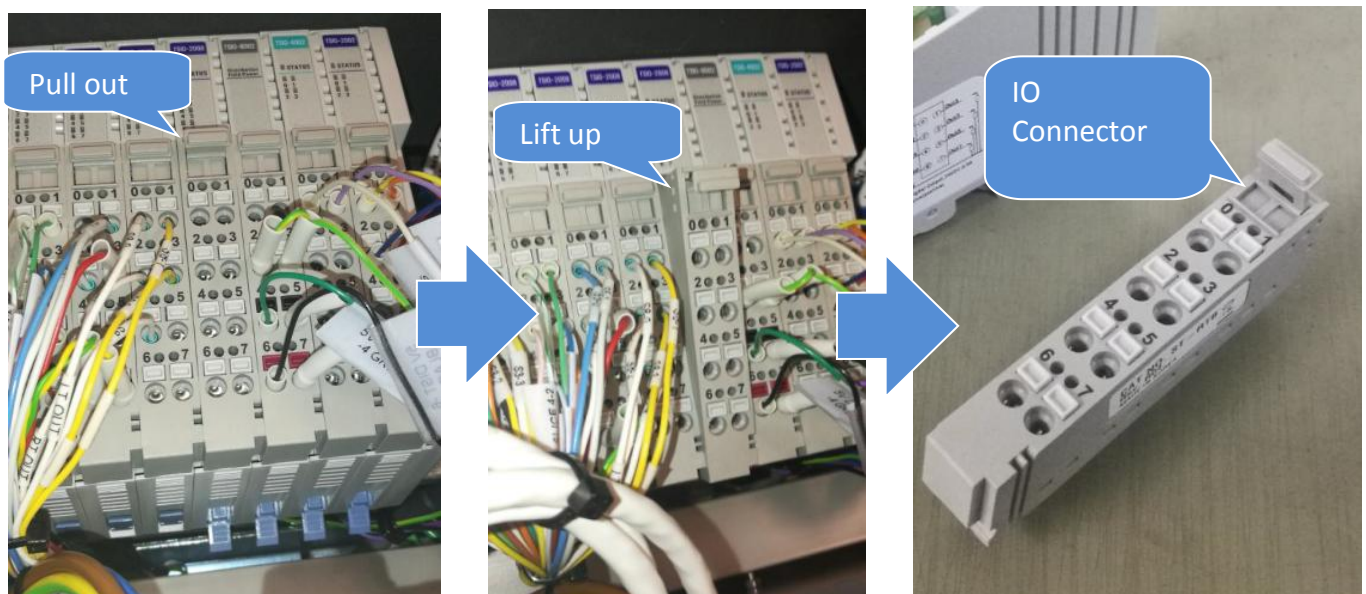


Figure 8 Slice IO Connector (Selected Slice IO)

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Replacement Procedure

10. AXI V810 S2EX machine DIO Signal Pin Layout is illustrated as below.

S2EX																											
7-7	7-6	6-7	6-6	6	7	9-7	9-6	8-7	8-6	5-7	5-6	4-7	4-6	3-7	3-6	10-7	10-6	2-7	2-6	1-7	1-6	0-7	0-6	6	7		
DIO In - 31	DIO In - 30	DIO Out - 31	DIO Out - 30			DIO In - 47	DIO In - 46	DIO In - 39	DIO In - 38	DIO In - 23	DIO In - 22	DIO In - 15	DIO In - 14	DIO In - 7	DIO In - 6	DIO Out - 39	DIO Out - 38	DIO Out - 23	DIO Out - 22	DIO Out - 15	DIO Out - 14	DIO Out - 7	DIO Out - 6				
				5V	5V			Optical PP ErrorSignal	Optical PP Done Signal	SMEBA Conveyor Panel Available	SMEBA Conveyor Not Ready	Interlock Service Mode On	Interlock Powered On	L Op-PP Sensor Engaged	R Op-PP Sensor Engaged		Reset Optical PP Sensor	SHEBA Unlatched Panel Out	SHEBA Good Panel Out	Z Motor Enable		Inner Barrier Control Valve	Bottom Clamp Up	24V	24V		
7-5	7-4	6-5	6-4	4	5	9-5	9-4	8-5	8-4	5-5	5-4	4-5	4-4	3-5	3-4	10-5	10-4	2-5	2-4	1-5	1-4	0-5	0-4	4	5		
DIO In - 29	DIO In - 28	DIO Out - 29	DIO Out - 28			DIO In - 45	DIO In - 44	DIO In - 37	DIO In - 36	DIO In - 21	DIO In - 20	DIO In - 13	DIO In - 12	DIO In - 5	DIO In - 4	DIO Out - 37	DIO Out - 36	DIO Out - 21	DIO Out - 20	DIO Out - 13	DIO Out - 12	DIO Out - 5	DIO Out - 4				
				0V	0V	PH-Motor Alarm	PH-Axis Position Mag #2	Optical PP Installed	Clamp Open	Right Filter Height Sensor	Left Filter Height Sensor	Interlock Logic Bit #5	Interlock Logic Bit #4	R Panel Clear Sensor Clear	L Panel Clear Sensor Clear	PH-Motor Bit #2	PH-Motor Bit #1	SHEBA Tester Panel Available	SHEBA Tester Not Ready			Belt Motor L/R	Belt Motor On/Off	0V	0V		
7-3	7-2	6-3	6-2	2	3	9-3	9-2	8-3	8-2	5-3	5-2	4-3	4-2	3-3	3-2	10-3	10-2	2-3	2-2	1-3	1-2	0-3	0-2	2	3		
DIO In - 27	DIO In - 26	DIO Out - 27	DIO Out - 26			DIO In - 43	DIO In - 42	DIO In - 35	DIO In - 34	DIO In - 19	DIO In - 18	DIO In - 11	DIO In - 10	DIO In - 3	DIO In - 2	DIO Out - 35	DIO Out - 34	DIO Out - 19	DIO Out - 18	DIO Out - 11	DIO Out - 10	DIO Out - 3	DIO Out - 2				
Cameras Controller Phase Locked	Cameras Not Using Excessive Current			GND	GND	PH-Axis Position Mag #1	Z-Axis Position Mag #3	Right OSES (PP) (Retract)	Right OSES (PP) (Extend)	Z-Motor Output (+)	Z-Axis Position Loading	Interlock Logic Bit #3	Interlock Logic Bit #2	Right Outer Barrier Open	Left Outer Barrier Open	PH-Motor Bit #0	PH-Motor Start	Z-Motor A+	Z-Motor B+	Light Tower Buzzer	Light Tower Green	Right OSES (PP) Extended	Left OSES (PP) Extended	GND	GND		
7-1	7-0	6-1	6-0	1	0	9-1	9-0	8-1	8-0	5-1	5-0	4-1	4-0	3-1	3-0	10-1	10-0	2-1	2-0	1-1	1-0	0-1	0-0	1	0		
DIO In - 25	DIO In - 24	DIO Out - 25	DIO Out - 24			DIO In - 41	DIO In - 40	DIO In - 33	DIO In - 32	DIO In - 17	DIO In - 16	DIO In - 9	DIO In - 8	DIO In - 1	DIO In - 0	DIO Out - 33	DIO Out - 32	DIO Out - 17	DIO Out - 16	DIO Out - 9	DIO Out - 8	DIO Out - 1	DIO Out - 0				
Camera Control Power On	Camera Control Cables Connected	Cameras Control Spare	Cameras Synch Clock			Z-Axis Position Mag #2	Z-Axis Position Mag #1	Left OSES (PP) (Retract)	Left OSES (PP) (Extend)	Inner Barrier Open	Inner Barrier Closed	Interlock Logic Bit #1	Interlock Logic Bit #0	Right Outer Barrier Closed	Left Outer Barrier Closed	PH-Motor Home	PH-Motor Alarm Reset	Enable Optical PP Controller	Panel Clamps Open	Light Tower Yellow	Light Tower Red	Right Outer Barrier Control Valve	Left Outer Barrier Control Valve	0V	24V		
Slice 7 - Input		Slice 6 - Output		Field Power Distributor		Slice 9 - Input		Slice 8 - Input		Slice 5 - Input		Slice 4 - Input		Slice 3 - Input		Slice 10 - Output		Slice 2 - Output		Slice 1 - Output		Slice 0 - Output		MES Slice IO Controller			

Figure 9 DIO Signal Pin Block Layout

11. Table 1 is indicate AXI S2EX machine Slide IO Part Number (Red Column at Figure 9)

No	Slide	Part Number	Description
1	MES Slice IO Controller	2150-0020	I/O, MODULE, SYNQNET NETWORK ADAPTER (TSIO-1001)
2	Slice 0 - Output	2150-0021	I/O, MODULE, 8-CHANNEL +24V OUTPUT (TSIO-4006)
3	Slice 1 - Output	2150-0021	I/O, MODULE, 8-CHANNEL +24V OUTPUT (TSIO-4006)
4	Slice 2 - Output	2150-0021	I/O, MODULE, 8-CHANNEL +24V OUTPUT (TSIO-4006)
5	Slice 3 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
6	Slice 4 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
7	Slice 5 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
8	Slice 8 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
9	Slice 9 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
10	Field Power Distributor	2140-0005	POWER SUPPLY, SILECE I/O DISTRIBUTOR (TSIO-8002)
11	Slice 6 - Output	2150-0023	I/O, MODULE, 4-CHANNEL +5V OUTPUT (TSIO-4002)
12	Slice 7 - Input	2150-0024	I/O, MODULE, 4-CHANNEL +5V INPUT (TSIO-2002)

Table 1 Slice I/O Part number

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I/O, MODULE, SYNQNET NETWORK ADAPTER Connector Replacement

12. Prepare I/O, MODULE, SYNQNET NETWORK ADAPTER (TSIO-1001) (P/N, 2150-0020) and remove the IO Connector as below Figure 10

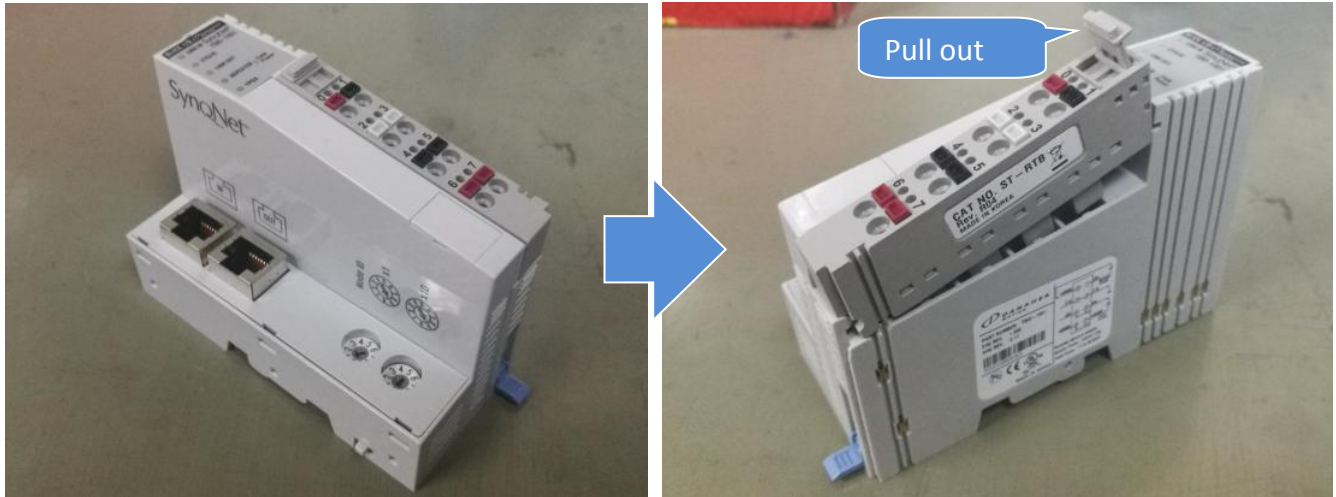


Figure 10 I/O, MODULE, SYNQNET NETWORK ADAPTER (TSIO-1001)

13. Replace NEW Slice IO Connector by refer to below orientation. Slot on the track & gently push in.

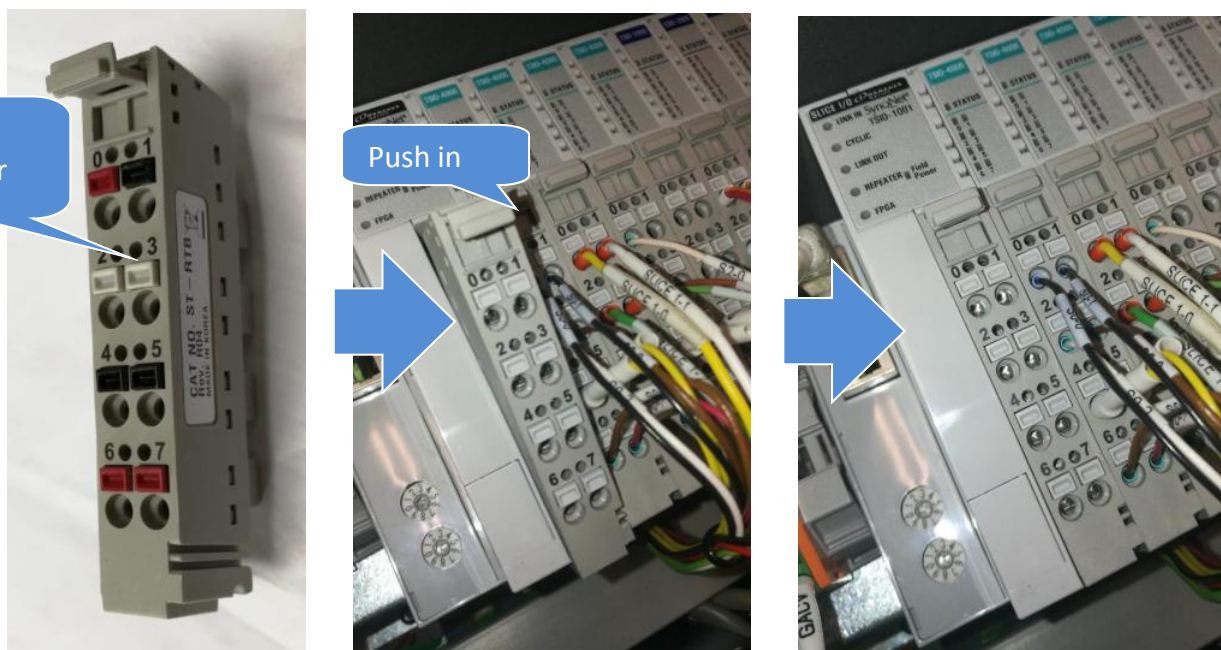


Figure 11

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14. While probe the Tiny Square Button, gently insert IO Cable respectively by refer to Figure 9

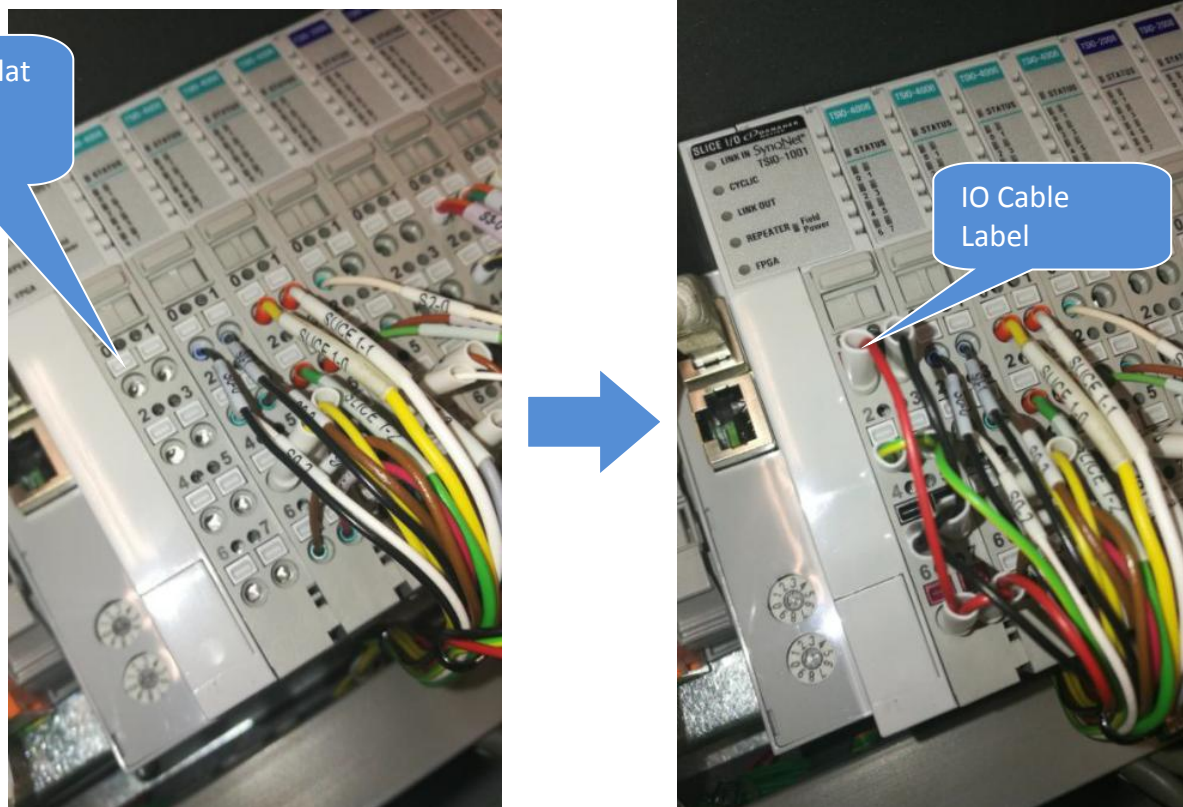


Figure 12 IO Cable Connection

Selected Slice IO Connector Replacement

15. Prepare Selected Slice IO by refer to Table 1. Remove IO Connector from Selected Slice IO

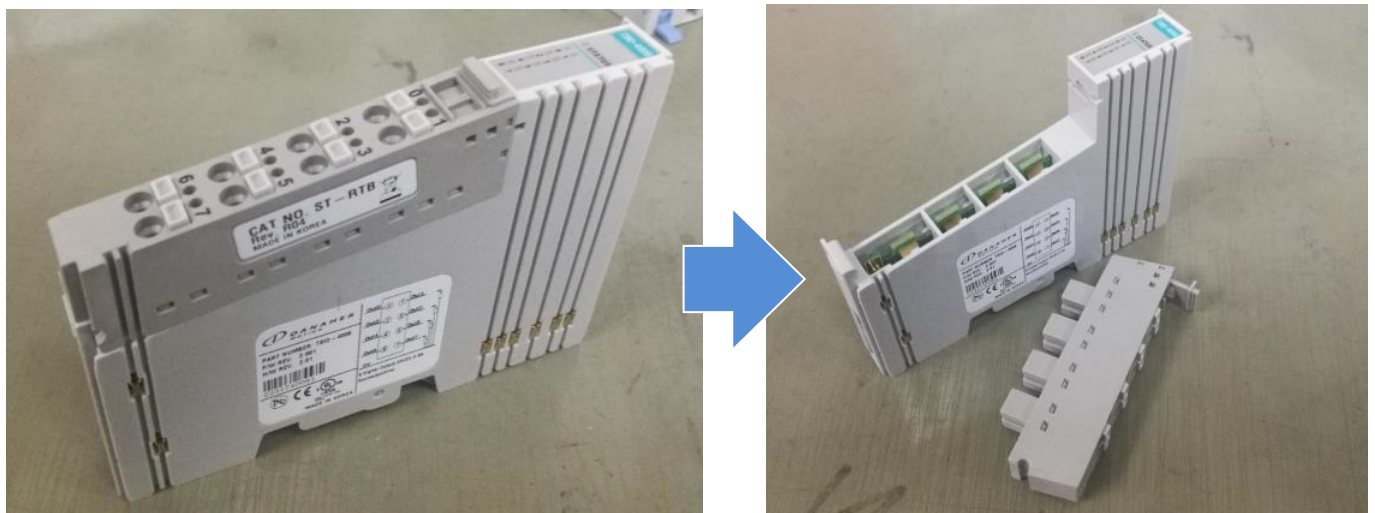


Figure 13 Selected Slice IO (Reference Picture Only)

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16. Replace NEW Selected Slice IO Connector by refer to below orientation. Slot on the track & gently push in.

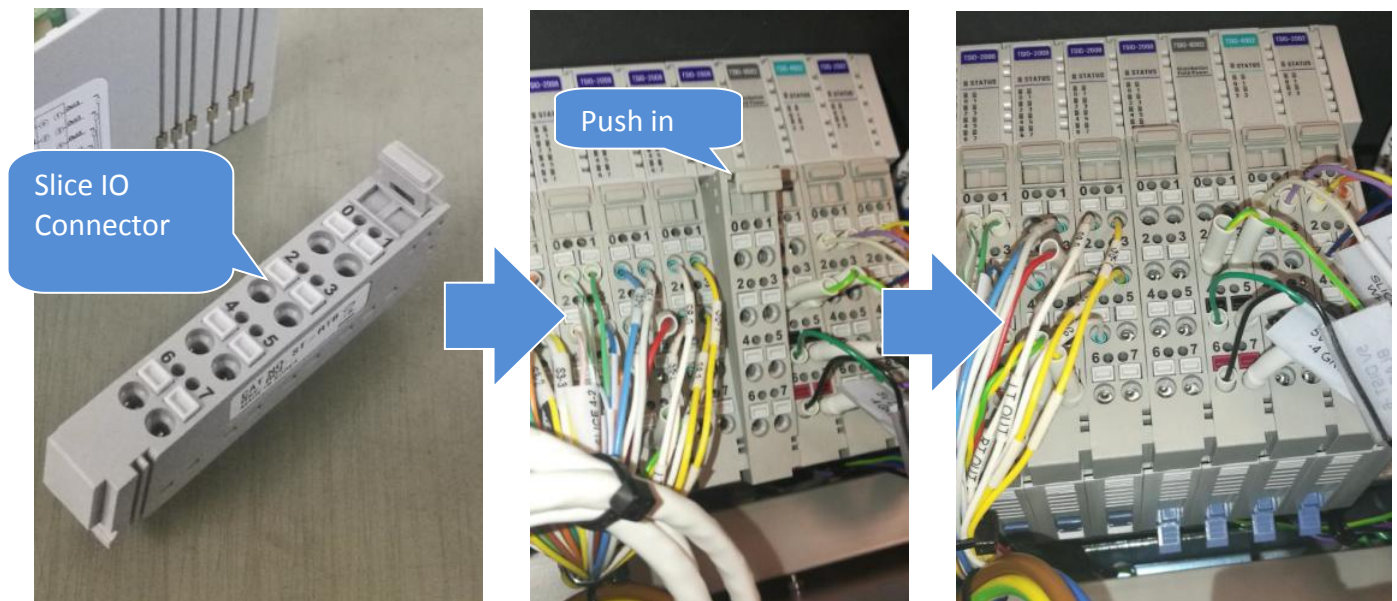


Figure 14 Selected Slice Connector Assembly

17. While probe the Tiny Square Button, gently insert IO Cable respectively by refer to Figure 9

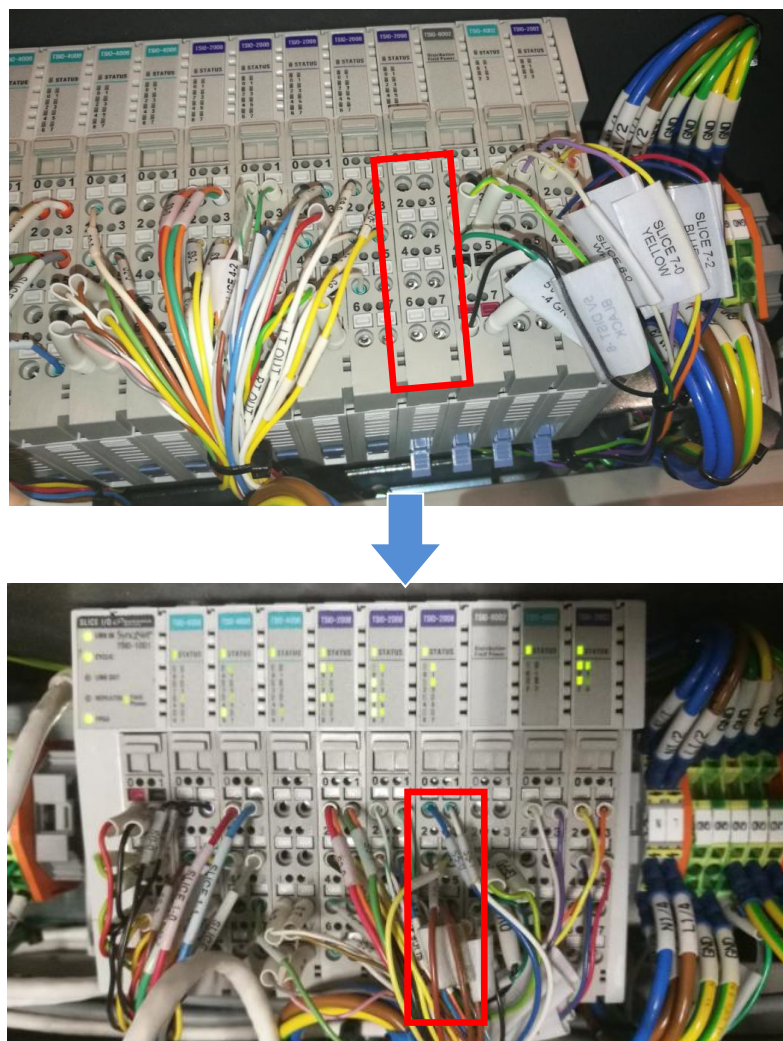


Figure 15 IO Cable Connection

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Verification Procedure

18. Use Multi-meter to **probe the 0vDC and 24vDC Terminal block**, Ohm reading should approximately **130 Ohm and above**

If Ohm Reading Close to Zero , Short Circuit potentially occur . Do Not Power on SSCA. Re-screen all cable connection by refer to Step 13 - 17

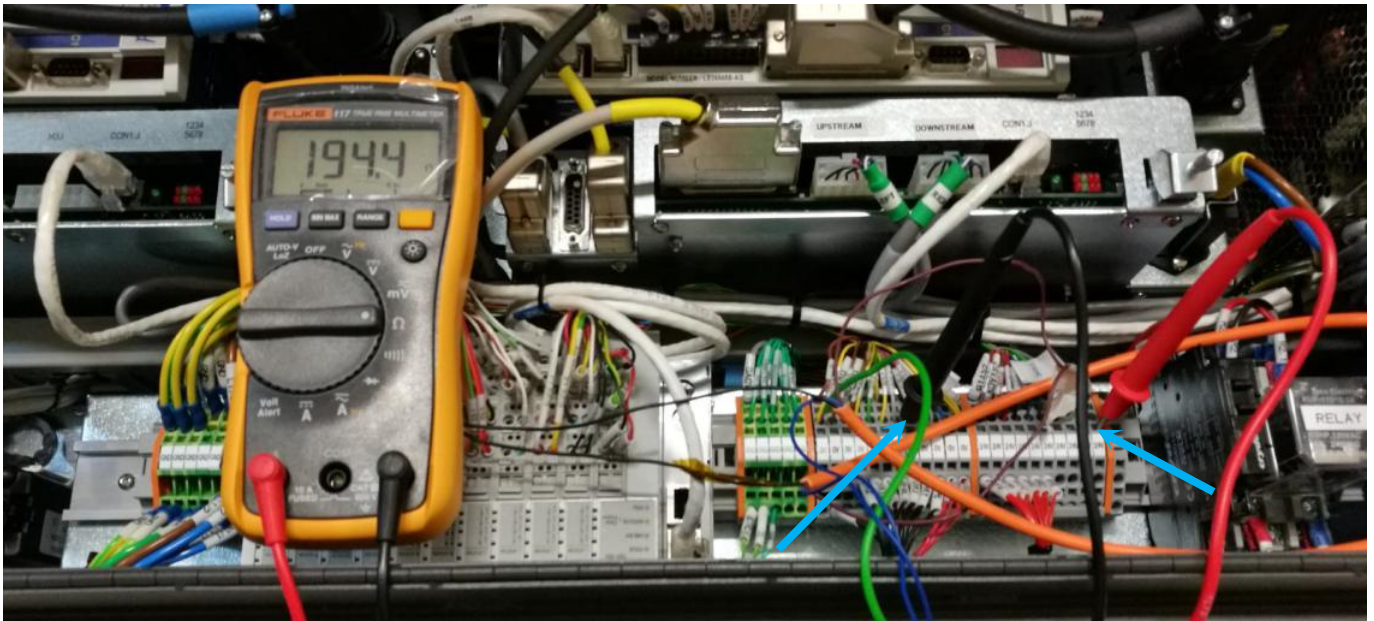


Figure 16 Terminal Block Verification

19. Open SSCA
20. Start-up system as usual
21. Run CDnA